

HIMANSHU GUPTA

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Professional Summary

Electronics & Communication Engineer with hands-on experience in solar PV design, process auditing, and embedded/IoT systems. Contributor to multiple published patent applications and a research paper in smart hardware solutions. Proficient in microcontrollers, circuit design, and system simulations, seeking opportunities in embedded systems and product development.

Experience

Apprentice – Renewable Energy / Solar PV

Pearce Service / Present

- Assisted in designing and optimizing solar PV layouts, SLDs, site plans, and electrical schematics using AutoCAD.
- Supported energy yield assessments and production simulations using PVsyst and Helioscope.
- Assisted with PV system design activities while following NEC/CEC requirements and international PV design standards.

Process Auditor

AVA Business Consulting & Risk Advisory LLP

- Conducted internal audits to verify compliance with organizational, regulatory, and statutory standards across multiple process areas.
- Identified process gaps and collaborated with cross-functional teams to implement workflow optimizations and risk mitigation strategies.

Embedded Systems & IoT Intern

Pantech Solutions / 2024

- Designed IoT circuits and programmed microcontrollers; integrated multi-sensor modules for automation prototypes using I2C, SPI, and UART protocols.
- Developed embedded projects using Raspberry Pi, ADC modules, and sensor fusion for real-time data acquisition.

Key Projects & Innovations

Bionic Arm Using EMG Sensors

Patent Filed

- Developed an EMG-based control system using MyoWare sensors and Raspberry Pi Pico for real-time muscle-driven servo actuation.

ShelterLink – IoT Smart Disaster Shelter

Patent No. 202411050202

- Built a sensor-based monitoring system for temperature, load, humidity, and occupancy with wireless transmission and real-time disaster alert notifications.

Autonomous Vehicle Prototype with Opti-Vision

Patent No. 202411078533

- Developed a Raspberry Pi-based embedded vision system with lane tracking, signal recognition, and obstacle detection using OpenCV.

Next-Gen Attendance Monitoring System

Published Research

- Deployed IR-based detection circuitry with Raspberry Pi for automated attendance logging and real-time dashboard analytics.

Technical Skills

Embedded Systems: Arduino, Raspberry Pi, Raspberry Pi Pico, ESP32, Microcontrollers

IoT & Communication: I2C, SPI, UART, Sensor Integration, Wireless Communication

Electronics & Circuit Design: Analog & Digital Electronics, Circuit Design, Sensors & Transducers, EMG Signal Acquisition, ADC/DAC

Programming & AI: C, Python, OpenCV, AI Prompt Engineering, Generative AI, AI-Assisted Research

Solar PV & CAD Design: AutoCAD (Site Plans, Solar PV Array Layouts, SLDs), Solar Pro Tool, Helioscope, Solar PV System Design

Simulation & Software Tools: Simulink, Circuit Simulation, MS Excel, Microsoft Office Suite, MS Visio

Education

B.E., Electronics & Communication Engineering

2021 – 2025

Chandigarh University

12th (Science), D.S.S.V. College, Simari, Bihar

2019 – 2021

Certifications & Languages

Certifications: Certified AppSec Practitioner | Cybersecurity Essentials | AWS Cloud Basics

Languages: Hindi (Native) | English (Professional)